TECHNICAL DILIGENCE REVIEW - RECORD 004

Helioscope Robotics, Inc. - diligence file reference TECHNICAL_DILIGENCE-004

1. Scope of the safety review and the documents examined

Pinehurst Systems Assurance was instructed by Ardent Peak to examine the functional-safety analysis, hazard register and protective-stop behaviour of the robot fleet; the resulting report is the report at TDR-2025-14, dated 2025-10-07. Pinehurst Systems Assurance delivered the present report on 2025-10-07; the scope agreed with Ardent Peak was the functional-safety analysis, hazard register and protective-stop behaviour of the robot fleet. The approach K. Vandermeer took in report TDR-2025-14 was to test the company's own assertions rather than to restate them. Method in this report was three-fold: read the design record, interview the owning engineers, and verify the claim against a running system. The reader of report TDR-2025-14 should treat each conclusion as bounded by the section it appears in.

K. Vandermeer tested a sample rather than the population in this report, and the sampling basis is stated in each section. K. Vandermeer was granted read access to the code, the pipeline and the incident tracker for the purposes of the report at TDR-2025-14; production access was not requested. Interviewees for the report at TDR-2025-14 were told that a stated uncertainty would be recorded as an uncertainty rather than resolved against them. Pinehurst Systems Assurance asked the same four questions of every team interviewed for this report, which is how the inconsistencies below were found. K. Vandermeer reports management's own position wherever it differs from the reviewer's, and does not merge the two.

10 follow-up items arose from the interview programme for report TDR-2025-14; 1 were still open at delivery. The interview programme behind TDR-2025-14 covered 5 teams and produced 10 follow-up requests, of which most were satisfied. TDR-2025-14 carries one structural limitation: the company's own metrics are the only source for several objectives, and the reviewer could not build an independent measurement inside the engagement. The principal limitation on the report at TDR-2025-14 is time: 5 weeks was enough to test the control path and not enough to test the reporting path. K. Vandermeer has resisted rounding the findings towards a comfortable conclusion in either direction.

One limitation of the report at TDR-2025-14 should be stated plainly: the reviewer had no production access and relied on the company's own measurements for live latency. K. Vandermeer did not visit the two most recently commissioned sites, so the commissioning quality there is taken from the company's records alone. This report separates matters that bear on the closing conditions from matters that belong in the first two quarters after close. K. Vandermeer would put 4 items from report TDR-2025-14 into the first hundred days rather than into a warranty. K. Vandermeer has separated what the company has built from what the company can demonstrate about what it has built.

The convention throughout TDR-2025-14 is that an absence of evidence is reported as such rather than as evidence of absence. Where this report says a defect is verified, the reviewer reproduced it or read a production trace of it; everything else is recorded as a concern. The transaction reading of the report at TDR-2025-14 is that the engineering risk is priced in the plan rather than in the purchase agreement. Deal implications in the present report are modest in aggregate and specific in detail: 2 items for the closing checklist, 4 for the plan. K. Vandermeer has kept the company's terminology even where a different word would have read better, so that a reader can search the source documents.

K. Vandermeer chose the sample for TDR-2025-14 after reading the dependency map, so the selection is not proportionate to headcount or to code volume. This report added two components to its sample after the first week, because the interview evidence pointed at them. Pinehurst Systems Assurance notes that 2 of the matters in the report at TDR-2025-14 are capable of being closed before completion if the company is asked in time. TDR-2025-14 recommends that 2 items be reflected in the disclosure schedules and that the remainder be tracked in the post-close plan. Where this report uses the word verified, it means the reviewer performed or witnessed something, and nothing weaker.

Schedule I - Interview programme and follow-up requests arising (TDR-2025-14)

Date	Participant	Role	Principal topic	Follow-ups raised	Follow-ups open
2025-09-15	Bettina Vogel	security engineer	Map Authoring Studio	5	2
2026-06-21	Tomas Villareal	quality lead	Localization Engine	4	3
2026-05-14	Colin Achebe	security engineer	Battery Management Service	5	0
2026-07-25	Yuki Harada	release manager	Traffic Manager	1	1

Date	Participant	Role	Principal topic	Follow-ups raised	Follow-ups open
2026-09-30	Marcus Duplessis	field reliability lead	WMS Connector	3	3
2025-09-14	Owen Brantley	head of engineering	Edge Gateway Agent	4	4
2026-06-01	Sam Oyelaran	engineering owner	WMS Connector	3	2
2026-03-27	Marta Kellerman	team lead	Fleet Allocator	5	0
2026-01-27	Tomas Villareal	team lead	Gripper Controller	1	1
2026-01-20	Daniel Okafor	quality lead	Video Snippet Service	3	2
2026-04-25	Farida Cisse	safety engineer	Config Distribution	4	3
2025-11-07	Erik Palmgren	quality lead	WMS Connector	1	1
2026-08-10	Yuki Harada	head of engineering	Fleet Allocator	1	0
2026-03-15	Rahul Menon	security engineer	Telemetry Ingest	2	1
2026-05-18	Grace Lindqvist	head of engineering	Payload Scale Interface	2	0
2025-09-01	Yuki Harada	security engineer	Payload Scale Interface	3	1
2026-03-13	Nadia Semenova	head of engineering	Safety Controller	1	1
2025-08-26	Wanda Njoku	site reliability lead	Fleet Health Monitor	4	4
2026-07-18	Farida Cisse	principal engineer	Speed and Separation Monitor	4	4
2025-09-18	Nadia Semenova	security engineer	Robot Shadow Store	5	3
2025-08-06	Anjali Sekhon	site reliability lead	Site Provisioning Service	3	2
2025-07-27	Daniel Okafor	safety engineer	Billing Usage Meter	2	1
2026-08-08	Yuki Harada	site reliability lead	Hardware-in-the-Loop Rig	2	1
2025-09-29	Wanda Njoku	safety engineer	OTA Updater	1	1
2025-12-01	Farida Cisse	quality lead	Battery Management Service	3	2
2026-02-09	Bettina Vogel	release manager	Wheel Odometry Filter	4	1
2026-05-11	Daniel Okafor	site reliability lead	Edge Gateway Agent	2	1
2026-03-28	Colin Achebe	principal engineer	Telemetry Ingest	3	2
2026-10-31	Sonia Krall	head of engineering	Video Snippet Service	2	0
2025-08-05	Bettina Vogel	safety engineer	Simulation Harness	4	4

2. Hazard identification: HZ-05 and the register as a whole

Register entry HZ-05 is assessed at Fatality severity with Improbable exposure and Low controllability. HZ-05 is mitigated by hardwired dual-channel stop chain independent of the compute module. The population exposed to hazard HZ-05 is the manipulator-equipped variant only. HZ-05 is reported at the evidence strength available and has not been promoted to fit the narrative.

Speed reduction engages too late for a fast-approaching person is the hazard entered at Serious injury hazard HZ-18. Register entry HZ-18 scores Serious injury on severity; exposure is Occasional and controllability Low. Where HZ-18 bites is the two pallet-mover variants only. HZ-18 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

The hazard recorded at register entry HZ-06 is that robot moves unexpectedly during maintenance. Occasional-exposure hazard HZ-06 depends for its mitigation on maintenance mode with a physical lockout and a two-step exit confirmation. On history, Occasional-exposure hazard HZ-06 produced a single event before the current control was fitted. Serious injury hazard HZ-06 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Equipment damage hazard HZ-07 produced a single event before the current control was fitted; that is worth stating alongside the assessed exposure. HZ-07 is the entry for the case that robot travels into an unmapped area and contacts fixed plant. Equipment damage hazard HZ-07 was scoped to robots fitted with the weighing deck, and the reviewer checked the scoping against the fleet record. register entry HZ-07 would move from a concern to a verified defect if one further measurement were supplied.

Post-mitigation, register entry HZ-08 sits at Tolerable and is recorded as verified. Severity at Equipment damage hazard HZ-08 is Equipment damage, with the exposure rated Frequent and controllability High. The manipulator-

equipped variant only is the exposed population for hazard HZ-08. hazard HZ-08 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Tolerable where cleaning compliance is evidenced is the residual position on Occasional-exposure hazard HZ-20, whose status is partially verified. Against HZ-20 the team relies on return-intensity monitoring with a cleaning prompt and a degraded-mode speed limit. HZ-20 was tested by contamination bench test plus a review of cleaning compliance at three sites, and the reviewer read the resulting record. hazard HZ-20 was inspected in the repository rather than described from the ticket, which is why the account is specific.

One qualification on Remote-exposure hazard HZ-04 the reviewer would not remove is that the gate landed in March 2026 after a field incident. The three risk terms on Serious injury hazard HZ-04 are severity Serious injury, exposure Remote and controllability Low. Remote-exposure hazard HZ-04 is mitigated by force limiting with a mechanical release and an interlock on the access panel. register entry HZ-04 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

The control applied at Serious injury hazard HZ-01 is speed and separation monitoring with a protective stop on margin breach. Evidence of verification at hazard HZ-01 consists of witnessed approach testing on six robots plus a review of the on-robot safety log. Register entry HZ-01 covers the case where robot contacts a pedestrian while travelling in a shared aisle. Serious injury hazard HZ-01 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Schedule N - Hazard identification and risk assessment register (TDR-2025-14)

Ref	Hazard	Severity	Exposure	Controllability	Mitigation	Verification method	Residual risk
HZ-01	Robot contacts a pedestrian while travelling in a shared aisle	Serious injury	Frequent	Low	speed and separation monitoring with a protective stop on margin breach	witnessed approach testing on six robots plus a review of the on-robot safety log	Tolerable with the measured approach speed assumption stated
HZ-02	Robot contacts a pedestrian emerging from a blind cross-aisle	Serious injury	Occasional	Low	aisle-end speed reduction plus a mirrored sensing field at mapped blind corners	site-specific commissioning test at each mapped blind corner	Tolerable at commissioned sites; two sites have unmapped corners
HZ-03	Load falls from an elevated deck during transport	Serious injury	Occasional	Medium	payload mass and centre-of-gravity limits enforced by the scale interface	bench drop testing plus a payload limit audit across the fitted fleet	Tolerable where the weighing deck is fitted
HZ-04	Gripper closes on a hand during a manual intervention	Serious injury	Remote	Low	force limiting with a mechanical release and an interlock on the access panel	force bench test after every calibration, evidenced by a calibration record	Tolerable once calibration is gated in the service workflow
HZ-05	Drive power is not removed when the emergency stop is actuated	Fatality	Improbable	Low	hardwired dual-channel stop chain independent of the compute module	witnessed actuation test on every unit at build and at annual service	Broadly acceptable
HZ-06	Robot moves unexpectedly during maintenance	Serious injury	Occasional	Low	maintenance mode with a physical lockout and a two-step exit confirmation	procedure walk-through with three service technicians and a lockout inventory	Tolerable
HZ-07	Robot travels into an unmapped area and contacts fixed plant	Equipment damage	Occasional	Medium	map keep-out enforcement plus a boundary confidence	simulation coverage of boundary cases plus a site survey audit	Tolerable where survey quality is confirmed

Ref	Hazard	Severity	Exposure	Controllability	Mitigation	Verification method	Residual risk
					check in localization		
HZ-08	Two robots collide at a corridor intersection	Equipment damage	Frequent	High	corridor reservation with deadlock detection and a deterministic yield tie-break	reservation protocol replay plus the pinch-point simulation scenario	Tolerable
HZ-09	Robot fails to stop on a descending ramp when a wheel loses traction	Serious injury	Remote	Medium	ramp speed limits, a slip detector and a mechanical brake hold	ramp testing on a wet and a dry surface at two sites	Not yet tolerable without the slip detector
HZ-10	Charging contact arcs during an unplanned disconnect	Fire	Improbable	Low	contactor sequencing with an arc-suppression design in the charge dock	supplier test report plus an internal witnessed disconnect test	Broadly acceptable
HZ-11	Battery pack thermal runaway during opportunity charging	Fire	Improbable	Low	cell temperature limits, pack-level current derating and a dock thermal cutout	supplier abuse test report plus a review of pack fault records	Broadly acceptable, subject to the state-of-health drift item
HZ-12	Operator is struck by a robot reversing out of a charge dock	Minor injury	Occasional	Low	reverse motion audible warning plus a rear sensing field	witnessed reverse test plus a review of the dock layout at four sites	Tolerable
HZ-13	Robot ignores a temporary barrier placed by site staff	Serious injury	Occasional	Medium	detection of unmapped obstacles with a stop, plus a site procedure for barriers	obstacle detection bench test plus a site procedure review	Tolerable where the barrier meets the visibility guidance
HZ-14	Remote operator issues a motion command to the wrong robot	Minor injury	Occasional	Low	explicit robot selection with a live video confirmation before motion	console interaction test plus a review of session records	Tolerable once the stale roster defect is closed
HZ-15	Robot obstructs an emergency egress route	Serious injury	Remote	Medium	egress routes marked as keep-out and a stop-and-clear behaviour on a fire signal	site commissioning check plus a fire drill observation at two sites	Tolerable
HZ-16	Firmware update leaves a robot in a mixed-version state	Equipment damage	Remote	Low	atomic update with rollback and a version consistency check before mission acceptance	two rollback drills plus a fleet version audit	Tolerable
HZ-17	Safety controller fault log overwrites evidence needed for an investigation	No direct harm	Frequent	Low	periodic fault log extraction to the cloud with a retention policy	log extraction record review across the fleet	Tolerable, with an evidence-loss caveat
HZ-18	Speed reduction engages too late for a fast-approaching person	Serious injury	Occasional	Low	protective distance calculation with a stated approach speed assumption	bench approach testing at the assumed speed and at a higher speed	Not yet tolerable at the higher approach speed

Ref	Hazard	Severity	Exposure	Controllability	Mitigation	Verification method	Residual risk
HZ-19	Robot enters a dock occupied by a person	Serious injury	Remote	Low	dock occupancy sensing plus a low approach speed inside the dock envelope	occupancy sensor bench test plus a witnessed dock approach	Tolerable
HZ-20	Lidar contamination degrades detection without a fault being raised	Serious injury	Occasional	Medium	return-intensity monitoring with a cleaning prompt and a degraded-mode speed limit	contamination bench test plus a review of cleaning compliance at three sites	Tolerable where cleaning compliance is evidenced
HZ-21	Manual push of a powered robot causes an uncontrolled movement	Minor injury	Remote	Low	push detection that engages the brake and requires a deliberate release	witnessed push test on two robot variants	Broadly acceptable
HZ-22	Site network loss leaves robots operating without fleet coordination	Equipment damage	Occasional	Medium	local yield behaviour with a conservative speed limit and a timed stop	network loss injection at one site plus a bench network partition test	Tolerable for outages under the tested duration

3. Risk assessment parameters and residual risk

The population exposed to hazard HZ-10 in the register is the newest three sites, where commissioning is incomplete. Fire harm, Improbable exposure and Low controllability are the parameters recorded for Improbable-exposure hazard HZ-10. Hazard HZ-10 produced a single event before the current control was fitted, which is the strongest available test of the control. Improbable-exposure hazard HZ-10 is reported at the evidence strength available and has not been promoted to fit the narrative.

In the reviewed period hazard HZ-11 in the register produced one field event that did not result in harm. The control applied at Fire hazard HZ-11 is cell temperature limits, pack-level current derating and a dock thermal cutout. Register entry HZ-11 identifies the hazard that battery pack thermal runaway during opportunity charging. register entry HZ-11 carries a follow-up request that was satisfied after the draft of TDR-2025-14 was circulated.

What the register does not say about hazard HZ-03 in the register, and the reviewer does, is that robots without the weighing deck rely on operator loading discipline. On the register's own scale, Occasional-exposure hazard HZ-03 is Serious injury severity, Occasional exposure, Medium controllability. Exposure under register entry HZ-03 extends to every robot variant currently in production. HZ-03 carries a follow-up request that was satisfied after the draft of TDR-2025-14 was circulated.

One caveat attaches to register entry HZ-17: extraction runs weekly, so a busy robot can still lose events between extractions. On history, register entry HZ-17 has produced no field event in the reviewed period. Post-mitigation, Frequent-exposure hazard HZ-17 sits at Tolerable, with an evidence-loss caveat and is recorded as open. HZ-17 is reported here in full because a summary of it in an earlier draft of TDR-2025-14 was misread as an assurance.

Exposure under hazard HZ-14 extends to the sites that have moved to opportunity charging. Entry hazard HZ-14 in the hazard register reads: remote operator issues a motion command to the wrong robot. One caveat attaches to Minor injury hazard HZ-14: the roster invalidation fix landed in July 2025. Occasional-exposure hazard HZ-14 was excluded from an earlier draft of TDR-2025-14 and reinstated after the reviewer obtained the supporting record.

The design measure recorded at HZ-21 is push detection that engages the brake and requires a deliberate release. The reviewer's note on Remote-exposure hazard HZ-21 is that both currently shipping variants were tested. Risk parameters for register entry HZ-21 are Minor injury severity, Remote exposure and Low controllability. hazard HZ-21 in the register is included because a reader of TDR-2025-14 should be able to disagree with the inference without repeating the work.

One caveat attaches to Serious injury hazard HZ-19: occupancy sensing is fitted at every dock revision currently deployed. Evidence of verification at hazard HZ-19 in the register consists of occupancy sensor bench test plus a witnessed dock approach. Serious injury hazard HZ-19 is relevant to the two pallet-mover variants only rather than to the whole fleet. hazard HZ-19 in the register is reported at the evidence strength available and has not been promoted to fit the narrative.

What the register does not say about hazard HZ-13 in the register, and the reviewer does, is that two sites use barriers below the guidance height and detection is then marginal. HZ-13 has produced no event, though the control has activated repeatedly, which is the strongest available test of the control. Serious injury hazard HZ-13 is addressed by detection of unmapped obstacles with a stop, plus a site procedure for barriers, and the reviewer traced that control into the design record. Serious injury hazard HZ-13 was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

4. Safety-related subsystems and their verification

The dependency map places the costmap fusion pipeline and the localization engine above Motion Planner and the traffic manager and the on-robot drive supervisor below it. Motion Planner owns its own data and exposes it only through a versioned interface, which the reviewer read. Test coverage on Motion Planner is 71 per cent line coverage and the measurement basis was confirmed with Priya Raghunathan. Of the 14 open items against Motion Planner, the reviewer treats one as material: replan oscillation when two robots negotiate the same narrow-aisle pinch point. Release cadence for Motion Planner is every three weeks, promoted ring by ring, and the release records bear that out. Motion Planner was excluded from an earlier draft of TDR-2025-14 and reinstated after the reviewer obtained the supporting record.

The published objective for Motion Planner is that plan cycle completes inside 40 ms at the 99th percentile. Underlying Motion Planner in this report are eleven design records, the planner benchmark suite and four weeks of production plan traces. The assessment of Motion Planner is not unqualified: there is no finding of unfitness; there is a finding that the team cannot currently demonstrate fitness. Motion Planner sits edge, one instance per robot with no cloud fallback, and no alternative placement is currently supported. Motion Planner was tested against the dependency ageing register, which records none; the toolchain is pinned to a supported compiler release. Motion Planner is included because a reader of TDR-2025-14 should be able to disagree with the inference without repeating the work.

TypeScript with a React front end is the implementation language for Remote Operations Console, which is hosted on cloud, a static bundle plus a session gateway with PostgreSQL 14 for session records and object storage for clips behind it. Remote Operations Console was assessed against the objective and the defect record, and no verified defect was identified, and two plausible concerns are recorded below at low evidence strength. Data ownership is clear for Remote Operations Console: it writes PostgreSQL 14 for session records and object storage for clips and no other service writes there. Upstream of Remote Operations Console sit the identity service, the telemetry ingest path and the video snippet service; downstream, the traffic manager and the alarm router consume its output. Remote Operations Console commits that operator input reaches the robot inside 350 ms at p95 on a supported network, and the objective is measured continuously rather than sampled. Remote Operations Console carries a follow-up request that was satisfied after the draft of TDR-2025-14 was circulated.

On paper Remote Operations Console releases twice a week, and the observed practice did not diverge from that. Against Remote Operations Console, the coverage figure is 52 per cent line coverage, which coverage alone overstates the assurance here, because the bench suite is the real safety net. 27 items sit in the Remote Operations Console backlog; the entry that bears on the investment case is that the session list keeps rendering a robot that has already left the site until a manual refresh. Responsibility for Remote Operations Console was confirmed with Anjali Sekhon, whose team, Applications, also holds the on-call duty. Remote Operations Console sits cloud, with a thin session relay in each region, and no alternative placement is currently supported. Remote Operations Console was raised again with the owning engineer after the initial interview, and the position did not change.

WMS Connector sits cloud, one deployment per customer tenant, and no alternative placement is currently supported. Integrations releases WMS Connector every two weeks, per tenant on request, and no release in the sampled window bypassed the gate set. Underlying WMS Connector in this report are six tenant integration specifications and a month of connector error logs. Read against its objective, WMS Connector management's own assessment is broadly

supported, with the coverage gap treated as a genuine finding rather than a presentational one. Integrations publishes one objective for WMS Connector: order acknowledgement returned to the warehouse system inside 1.5 s at p95. WMS Connector was tested rather than accepted, and the test is described in the working papers behind TDR-2025-14.

Tomas Villareal leads Integrations's work on WMS Connector and was the reviewer's principal interview for it. WMS Connector carries 23 open defects, among them a case where duplicate order suppression relies on a customer-supplied identifier that two tenants reuse. WMS Connector does not share a schema with any consumer; every read by the order broker and the billing usage meter goes through the published interface. End-of-life review of WMS Connector found one customer integration still runs on a connector build from 2024. The dependency map places the customer warehouse management systems and the identity service above WMS Connector and the order broker and the billing usage meter below it. WMS Connector is the clearest instance in TDR-2025-14 of engineering practice being better than the documentation suggests.

The dependency map places the edge gateway agent and the event bus above Telemetry Ingest and the fleet health monitor, the analytics warehouse loader and the alarm router below it. Telemetry Ingest carries 19 open defects, among them a case where the edge buffer drops the oldest samples after four hours offline without raising an alert. What the reviewer had for Telemetry Ingest was the telemetry schema registry, retention configuration and a forced-offline buffer test; the conclusions below do not reach past it. Telemetry Ingest ships weekly, with schema changes gated on a compatibility check; the reviewer sampled the last twelve releases. The reviewer's judgement on Telemetry Ingest is that the reviewer accepts the engineering position while noting that the verifying test does not exist. Telemetry Ingest was reviewed against the position twelve months earlier, and nothing measurable has changed.

End-of-life review of Telemetry Ingest found the stream processor runs a release that leaves community support in late 2026. One coupling concern applies to Telemetry Ingest: a time-series store with ninety-day retention and object storage beyond that is shared with a second service, so a schema change is not a local decision. Deployment of Telemetry Ingest is cloud, one ingest endpoint per region with an edge buffer on the robot, which sets the blast radius of any defect in it. The published objective for Telemetry Ingest is that telemetry visible in the operator console inside 30 s of emission at p95. Telemetry Ingest pairs a Go 1.22 with a Rust decoder implementation with cloud, containers plus a stream processor, and its data lives in a time-series store with ninety-day retention and object storage beyond that. Telemetry Ingest is the reason one section of TDR-2025-14 concludes narrowly rather than generally.

The evidence base for Video Snippet Service was the retention configuration, a lifecycle audit across regions and the access log sample. Of the 12 open items against Video Snippet Service, the reviewer treats one as material: the retention lifecycle was applied per bucket, so one region kept clips for ninety days. Applications releases Video Snippet Service monthly, and no release in the sampled window bypassed the gate set. Video Snippet Service pairs a Go 1.22 with a media pipeline implementation with cloud, containers plus a media transcoder, and its data lives in object storage with a thirty-day lifecycle and a PostgreSQL 14 index. In production Video Snippet Service is cloud, regional, with an on-robot capture buffer, unchanged since the last architecture review. Video Snippet Service was compared with the equivalent position at two other components, and it is neither the best nor the worst.

For Video Snippet Service the reviewer records that no verified defect was identified, and two plausible concerns are recorded below at low evidence strength. Upstream of Video Snippet Service sit the on-robot camera buffer and the identity service; downstream, the remote operations console and the incident review workflow consume its output. Against Video Snippet Service, the coverage figure is 43 per cent line coverage, which the figure is respectable but excludes the fallback path, which is where the known defect sits. The interface Video Snippet Service publishes is versioned, and the remote operations console and the incident review workflow were checked for use of any undocumented field. Video Snippet Service commits that a requested snippet is available to the operator inside 45 s, and the objective is measured continuously rather than sampled. Video Snippet Service is the reason one section of TDR-2025-14 concludes narrowly rather than generally.

Billing Usage Meter owns its own data and exposes it only through a versioned interface, which the reviewer read. In production Billing Usage Meter is cloud, regional, unchanged since the last architecture review. Against Billing Usage Meter, the coverage figure is 79 per cent line coverage, which the reviewer notes that the untested share includes the error handling. Hosting for Billing Usage Meter is cloud, containers on a managed Kubernetes cluster, and its Go 1.22 code writes through to PostgreSQL 14 with an immutable usage ledger. Billing Usage Meter releases monthly, aligned to the

billing cycle. Billing Usage Meter is included because a reader of TDR-2025-14 should be able to disagree with the inference without repeating the work.

The published objective for Billing Usage Meter is that metered usage reconciles to the order journal within 0.5 per cent each month. The evidence base for Billing Usage Meter was the metering specification, three monthly reconciliations and the ledger immutability check. The finance export and the customer usage report rely on Billing Usage Meter, which in turn relies on the order broker and the WMS connector. Billing Usage Meter has a single named owner, Nadia Semenova of Data and Telemetry, which is not true of every service reviewed. Billing Usage Meter's dependency ageing was checked: none identified. Billing Usage Meter is one of the entries where TDR-2025-14 relies on the company's own measurement rather than an independent one.

Lidar Driver publishes a stable interface, though two fields used by the costmap fusion pipeline and the localization engine are documented only in code comments. Lidar Driver is promoted as needed, historically twice a year, with the ring order fixed by policy. Robot Firmware tracks 18 open defects on Lidar Driver, including the acknowledged case where reconnect handling leaves the timestamp offset uncorrected for the first three frames. For Lidar Driver the reviewer worked from the driver source, the vendor integration guide and a captured reconnect trace. Robot Firmware holds Lidar Driver, and the ownership entry naming Yuki Harada matches the on-call rotation. Lidar Driver was checked against the corresponding entry in the company's own register, and the register was current.

The dependency map places the sensor hardware and the time synchronisation daemon above Lidar Driver and the costmap fusion pipeline and the localization engine below it. End-of-life review of Lidar Driver found the vendor's software development kit is two major versions behind current. No store; frames are handed to fusion in memory is the system of record for Lidar Driver, a C++17 component running on on-robot Linux, user-space driver. 44 per cent line coverage is reported for Lidar Driver, and the reviewer re-ran the measurement on a clean checkout. The assessment of Lidar Driver is not unqualified: there is no finding of unfitness; there is a finding that the team cannot currently demonstrate fitness. Lidar Driver was one of two items where TDR-2025-14 could not obtain the underlying data within the engagement.

Schedule A - Service inventory, ownership and objective attainment (TDR-2025-14)

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
Motion Planner	Rust 1.79	Autonomy Core	71%	plan cycle completes inside 40 ms at the 99th percentile	the costmap fusion pipeline and the localization engine	none; the toolchain is pinned to a supported compiler release
Costmap Fusion	C++17	Perception	58%	fused occupancy grid published at 20 Hz with sensor-to-grid latency under 60 ms	the lidar driver, the depth camera bridge and the wheel odometry filter	a vendored geometry library whose upstream branch left support in 2024
Fleet Allocator	Go 1.22	Fleet Services	82%	task assignment inside 900 ms of order arrival at p95	the order broker and the robot shadow store	none identified in the reviewed manifests
Traffic Manager	Go 1.22	Fleet Services	74%	corridor deadlock detected and broken inside two seconds	the fleet allocator and the site map service	Redis 6 client library still vendored in one code path
Charge Scheduler	Python 3.11	Fleet Services	63%	fleet state of charge stays above 35 per cent through the peak shift at every site	the battery management service and the site shift calendar	none; the dependency set was refreshed in the 2026 platform upgrade
Safety Controller	C, written to a constrained coding subset	Safety Engineering	91%	a validated protective-stop trigger de-energises the drive inside 120 ms	the E-stop chain, the speed and separation monitor and the drive current sensors	none; the supplier's support commitment runs to 2031
E-Stop Chain	VHDL	Safety Engineering	witnessed test	drive power is removed within	the operator stop buttons and the	none; the relay parts are current

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
				250 ms of button actuation on every tested unit	perimeter interlocks	production
Lidar Driver	C++17	Robot Firmware	44%	no more than one dropped scan per ten thousand at nominal spin rate	the sensor hardware and the time synchronisation daemon	the vendor's software development kit is two major versions behind current
Localization Engine	C++17	Autonomy Core	66%	pose error stays under 25 mm in mapped aisles at survey speed	the lidar driver, the wheel odometry filter and the map distribution channel	an unmaintained scan-matching library is still linked in the fallback path
WMS Connector	Java 21	Integrations	69%	order acknowledgement returned to the warehouse system inside 1.5 s at p95	the customer warehouse management systems and the identity service	one customer integration still runs on a connector build from 2024
Order Broker	Go 1.22	Fleet Services	78%	no accepted order is lost, measured by journal reconciliation each night	the WMS connector and the site configuration service	none identified
Telemetry Ingest	Go 1.22	Data and Telemetry	61%	telemetry visible in the operator console inside 30 s of emission at p95	the edge gateway agent and the event bus	the stream processor runs a release that leaves community support in late 2026
OTA Updater	Rust 1.79 on the robot, Go 1.22 in the cloud	Platform Infrastructure	76%	an update is either fully applied or fully rolled back, with no mixed-version robot	the release orchestrator and the artefact signing service	none; artefact signing was moved to the current key format in 2025
Identity and Authorization	Go 1.22	Security Engineering	84%	token issuance inside 200 ms at p95 and no unauthenticated write path	the customer directory federations and the secret store	none identified in the reviewed manifests
Remote Operations Console	TypeScript	Applications	52%	operator input reaches the robot inside 350 ms at p95 on a supported network	the identity service, the telemetry ingest path and the video snippet service	a charting library that no longer receives security updates
Simulation Harness	Python 3.11	Developer Productivity	47%	the nightly scenario suite completes inside four hours	the planner and localization builds under test	the physics core depends on a library branch that stopped receiving fixes in 2023
Hardware-in-the-Loop Rig	Python 3.11 test harness around production firmware	Robot Firmware	39%	every firmware candidate runs the full bench suite before ring one	the firmware candidate artefacts and the bench sensor stimulators	the rig controller runs an operating system release that left vendor support in 2025

4. Safety-related subsystems and their verification - part 2 of 2

Where the 4. Safety-related subsystems and their verification was interrupted by the page, it picks up again below, and the numbering is continuous throughout the schedule. The sequence is unbroken either side of it.

Treat the conventions declared for the 4. Safety-related subsystems and their verification as governing here too; this part introduces none of its own. The break is editorial.

Schedule A - Service inventory, ownership and objective attainment (TDR-2025-14) (continued, part 2 of 2)

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
Fleet Health Monitor	Python 3.11	Field Reliability	57%	a fleet-affecting condition raises an alert inside three minutes of first symptom	the telemetry ingest path and the robot shadow store	none identified
Battery Management Service	C on the pack controller, Python 3.11 in the cloud	Robot Firmware	54%	pack cell imbalance is reported before it reaches the derating threshold	the pack sensors and the charge scheduler	one pack revision uses a cell supplier that has since discontinued the part
Gripper Controller	C++17	Robot Firmware	41%	a grasp completes or reports failure inside 1.2 s	the payload scale interface and the perception grasp proposals	none; the coprocessor part is current production
Map Authoring Studio	TypeScript	Applications	49%	a published map revision reaches every robot at the site inside fifteen minutes	the site survey imports and the identity service	a geometry rendering library one major version behind its supported line
Site Provisioning Service	Go 1.22	Platform Infrastructure	64%	a new site reaches a working configuration inside one working day	the identity service and the customer network intake records	none identified
Event Bus	operated open-source broker, no in-house code	Platform Infrastructure	witnessed test	publish acknowledgement inside 25 ms at p95 and no partition without a leader	every publishing service	the broker minor release in production leaves the vendor's support window in 2027
Config Distribution	Rust 1.79 agent	Platform Infrastructure	72%	a configuration change reaches every online robot inside five minutes	the site provisioning service and the artefact signing service	none; the bundle format was revised in 2025
Robot Shadow Store	Go 1.22	Fleet Services	68%	shadow state is no more than two seconds behind the robot at p95	the telemetry ingest path and the edge gateway agent	none identified
Analytics Warehouse Loader	Python 3.11	Data and Telemetry	45%	the daily load completes before 06:00 local time at every region	the telemetry ingest path, the order broker and the billing usage meter	one transformation still runs on a deprecated warehouse function
Billing Usage Meter	Go 1.22	Data and Telemetry	79%	metered usage reconciles to the order journal within 0.5 per cent each month	the order broker and the WMS connector	none identified
Alarm Router	Python 3.11	Field Reliability	51%	a dispatched alarm reaches an on-call responder inside 90 s at p95	the fleet health monitor and the on-call rotation source	the paging provider's older application interface is still in use
Video Snippet Service	Go 1.22	Applications	43%	a requested snippet is available to the operator inside 45 s	the on-robot camera buffer and the identity service	a transcoder build with known but low-severity upstream advisories
Speed and Separation Monitor	C++17	Safety Engineering	87%	commanded speed is reduced before the separation distance falls below the protective distance	the costmap fusion pipeline and the safety-rated scanner inputs	none; the sensing hardware is current production

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
Wheel Odometry Filter	C++17	Robot Firmware	62%	odometry drift stays under 1.5 per cent of distance travelled on a clean floor	the wheel encoders and the inertial measurement unit	none identified
Payload Scale Interface	C++17	Robot Firmware	36%	payload mass is reported within 2 per cent of the reference load	the load cell hardware	none identified
Edge Gateway Agent	Rust 1.79	Platform Infrastructure	74%	no telemetry is lost for an outage shorter than four hours	the on-robot services that emit telemetry	none; the transport library was updated in early 2026
Release Orchestrator	Go 1.22	Developer Productivity	55%	no artefact reaches production without a recorded approval and a passing gate set	the build system, the simulation harness gate and the identity service	none identified

5. Safety-relevant incidents and protective-stop behaviour

On delivery, incident INC-2026-015 shows this: landed on 2026-03-03. Reading Config Distribution incident INC-2026-015 from its first entry, on 2026-02-09 a robot that had been offline at publication applied a configuration bundle without re-validating preconditions. Fathom Mobility felt INC-2026-015, which affected 2 sites and 7 robots. Strong on the cause and limited on the contributing factor is the evidence grade for Fathom Mobility incident INC-2026-015, and the reader should not read severity into it. Validation event INC-2026-015 produced one commitment: to move precondition validation to apply time and add an offline apply case to the platform suite, owned by Sam Oyelaran, dated 2026-03-31. validation event INC-2026-015 rests on a single source, and the reviewer has said so rather than implying corroboration.

INC-2026-015 was caused by a design gap: precondition validation ran at publication time rather than at apply time. Detection came through the on-robot supervisor raised a validation fault on the next mission, which places validation incident INC-2026-015 on the reported side of the ledger. Mitigation of Fathom Mobility incident INC-2026-015 took 44 m and full resolution took 22 days. The reviewer treats one contributing factor in Fathom Mobility incident INC-2026-015 as the more important lesson: the offline apply path was added for a customer with intermittent connectivity and was lightly reviewed. Incident INC-2026-015 is not read as an isolated failure: the same class of defect can recur in two sibling components that were not changed. INC-2026-015 was one of two items where TDR-2025-14 could not obtain the underlying data within the engagement.

Speed and Separation Monitor incident INC-2026-033 carries a second-order finding: the assumption was documented in the internal safety case and accepted at the time. Safety margin event INC-2026-033 affected one site and 6 robots. Incident INC-2026-033 shows a short mitigation and a long tail: immediate speed reduction; 3 h to review the fleet against open at the review date. The post-mortem for Speed and Separation Monitor incident INC-2026-033 identifies the cause as follows: the protective distance uses a fixed assumed approach speed rather than a measured closing rate. For incident INC-2026-033 the follow-through was tested rather than assumed, and not landed; a design study is running and the vendor assessment is booked. incident INC-2026-033 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

Speed and Separation Monitor owns Speed and Separation Monitor incident INC-2026-033, which the reviewer classes as safety margin. Asterion Logistics incident INC-2026-033 reached the engineering team when the on-robot safety log flagged a margin breach that did not become a contact event. The immediate event in safety margin incident INC-2026-033 was that a speed reduction engaged later than expected when an operator approached faster than the assumed walking speed, dated 2026-03-25. Grading INC-2026-033 on evidence rather than on consequence gives moderate; the narrative is consistent but the reviewer could not independently reproduce it. Asterion Logistics incident INC-2026-033 produced one commitment: to measure closing rate where the sensing permits and re-run the protective

distance calculation, owned by Bettina Vogel, dated 2026-08-31. INC-2026-033 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Classifying Motion Planner incident INC-2025-103 as algorithm defect matters because the Motion Planner backlog contains two further items of the same kind. Motion Planner incident INC-2025-103 records that on 2025-08-22, two robots repeatedly yielded to each other in a narrow aisle and both stalled for nineteen minutes. 19 m elapsed before algorithm defect incident INC-2025-103 was mitigated, with resolution at 58 days. Verification of the algorithm defect event INC-2025-103 remediation gives the tie-break landed on 2025-10-19; the scenario was reinstated on 2025-09-02. Detection came through the queue depth alert introduced after the March incident fired correctly, which places INC-2025-103 on the alerted side of the ledger. Ironwood Grocery Group incident INC-2025-103 was cross-checked against the engineering record supplied for TDR-2025-14, and the two agree.

Algorithm defect incident INC-2025-103 would have been shorter or smaller had it not been that the simulation scenario for this geometry existed but had been excluded from the nightly suite. Moderate, resting on the team's own post-mortem and one supporting log extract is the evidence grade for algorithm defect incident INC-2025-103, and the reader should not read severity into it. The engineering team's root cause for incident INC-2025-103 is that the yield heuristic had no tie-break and both robots computed the same deference score, and the reviewer found the analysis sound. The blast radius of Ironwood Grocery Group incident INC-2025-103 was one site and 2 robots. The reviewer's reading of Ironwood Grocery Group incident INC-2025-103 is that the team's analysis was honest about a gap it could have concealed. Ironwood Grocery Group incident INC-2025-103 carries a follow-up request that was satisfied after the draft of TDR-2025-14 was circulated.

Battery Management Service owns Battery Management Service incident INC-2026-057, which the reviewer classes as modelling. Modelling event INC-2026-057 would have been shorter or smaller had it not been that the two affected sites moved to opportunity charging after the estimator had been validated. Reading modelling incident INC-2026-057 from its first entry, on 2026-05-19 state-of-health estimates on opportunity-charged packs drifted enough to hide a cell imbalance until derating. Incident INC-2026-057 carries limited; the log retention window had closed before the reviewer asked for the raw data evidence, and the reviewer declined to raise the finding beyond what that supports. The corrective action for Dunbar Industrial incident INC-2026-057 is to refit the estimator on opportunity-charge data and add an imbalance alert independent of state of health, and 2026-09-30 is the date the team gave. incident INC-2026-057 is the clearest instance in TDR-2025-14 of a control that exists on paper and is weaker in practice.

Modelling event INC-2026-057 supports one conclusion and not another: the customer-reported detection path is the finding, not the defect itself. Modelling incident INC-2026-057 touched 2 sites and 5 robots, and Dunbar Industrial was the affected account. The detection path for Dunbar Industrial incident INC-2026-057 was a robot derated mid-shift and the pack was pulled for analysis. As to whether the Dunbar Industrial incident INC-2026-057 remediation actually landed: not landed; data collection is running and the imbalance alert is in review. The engineering team's root cause for Dunbar Industrial incident INC-2026-057 is that the estimator was fitted on packs that follow a full charge cycle, and the reviewer found the analysis sound. modelling event INC-2026-057 is the reason one section of TDR-2025-14 concludes narrowly rather than generally.

Wanda Njoku committed after incident INC-2025-096 to replace the mailbox fallback with a secondary responder and import rotations from the scheduling system, due 2025-10-15. What went wrong in Alarm Router incident INC-2025-096 was that the routing table fell back to a mailbox rather than to a secondary responder. Not applicable; no service impact elapsed before INC-2025-096 was mitigated, with resolution at 26 days. The classification applied to Alarm Router incident INC-2025-096 is process gap, and the affected component is Alarm Router. The affected population in incident INC-2025-096 was no production site and no robot directly, the exposure being to internal systems. INC-2025-096 was excluded from an earlier draft of TDR-2025-14 and reinstated after the reviewer obtained the supporting record.

The judgement recorded against incident INC-2025-096 is that management's severity rating is defensible and its remediation priority is not. Internal incident INC-2025-096 rests on strong, since the reviewer reproduced the failure mode in a laboratory environment evidence, which the reviewer states separately from the severity of the finding. The immediate event in Alarm Router incident INC-2025-096 was that an overnight alarm routed to a shared mailbox because the on-call rotation had a two-hour gap, dated 2025-08-05. On delivery, incident INC-2025-096 shows this: the fallback change landed on 2025-08-31; the rotation import was deferred to 2026. A contributing factor in process gap incident

INC-2025-096 was that the rotation source of truth was a spreadsheet maintained outside the alarm router. incident INC-2025-096 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

A contributing factor in performance incident INC-2026-004 was that the site was surveyed with a sparse feature set that increases cache misses. Cause analysis in performance incident INC-2026-004 concluded that the fallback scan matcher is slower than the primary and is selected whenever the map cache misses. Moderate; the narrative is consistent but the reviewer could not independently reproduce it is the evidence grade for Kestrel Medical Supply incident INC-2026-004, and the reader should not read severity into it. Mitigation of performance incident INC-2026-004 took not a service outage; 5 days to a workaround and full resolution took open at the review date. Detection came through the site's commissioning engineer raised it during acceptance testing, which places INC-2026-004 on the reported side of the ledger. Localization Engine incident INC-2026-004 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Incident INC-2026-004 produced one commitment: to replace the fallback matcher and re-survey the affected aisles, owned by Rahul Menon, dated 2026-06-30. The affected population in Kestrel Medical Supply incident INC-2026-004 was one site and 18 robots. Performance incident INC-2026-004 records that on 2026-01-13, relocalisation after a lift-and-place exceeded the operator prompt timeout at a newly surveyed site. Verification of the performance incident INC-2026-004 remediation gives not landed; the matcher replacement is in design and the re-survey is scheduled. Localization Engine accounts for Localization Engine incident INC-2026-004, a performance event. INC-2026-004 was reviewed against the position twelve months earlier, and nothing measurable has changed.

Assessed on its own record, INC-2025-139 shows that the incident is unremarkable in itself and revealing about the observability of the surrounding system. The classification applied to multiple sites incident INC-2025-139 is capacity, and the affected component is Event Bus. Grading incident INC-2025-139 on evidence rather than on consequence gives moderate on the cause and strong on the delivery position, which was checked in the repository. As to whether the capacity event INC-2025-139 remediation actually landed: the split landed on 2026-01-04; the quarterly review has not yet run. Event Bus incident INC-2025-139 carries a second-order finding: the topic inventory had recorded the sharing as acceptable in 2023 and it was never revisited. capacity incident INC-2025-139 is included because a reader of TDR-2025-14 should be able to disagree with the inference without repeating the work.

Erik Palmgren committed after multiple sites incident INC-2025-139 to split control traffic onto a dedicated topic with its own quota and revisit the topic inventory quarterly, due 2026-01-31. What went wrong in capacity event INC-2025-139 was that control and telemetry traffic shared one topic and one consumer group budget. The immediate event in capacity incident INC-2025-139 was that a telemetry backlog on a shared topic delayed control events by up to eleven seconds, dated 2025-11-02. Time to mitigate Event Bus incident INC-2025-139 was 2 h 47 m, and time to resolve was 63 days. Capacity incident INC-2025-139 reached the engineering team when the traffic manager's own latency alert fired first. incident INC-2025-139 is one of the entries where TDR-2025-14 relies on the company's own measurement rather than an independent one.

The two durations recorded for resource leak event INC-2025-041 are 1 h 52 m to mitigate and 9 days to resolve. Grading resource leak event INC-2025-041 on evidence rather than on consequence gives moderate; the narrative is consistent but the reviewer could not independently reproduce it. Traffic Manager incident INC-2025-041 affected 2 sites and 61 robots. 2025-03-14 is the date of resource leak incident INC-2025-041, on which a robot lost its network link mid-traverse and its corridor reservation was never released. Verification of the Crestline Fulfillment incident INC-2025-041 remediation gives landed on 2025-05-22, verified in the reservation replay suite. Crestline Fulfillment incident INC-2025-041 is treated as material to the investment case rather than as a routine engineering item.

What went wrong in incident INC-2025-041 was that the reservation lease had no expiry, so release depended on the departing robot. How resource leak event INC-2025-041 surfaced matters to the assessment: a site supervisor telephoned the on-call engineer after robots queued at one intersection. Traffic Manager owns resource leak incident INC-2025-041, which the reviewer classes as resource leak. Crestline Fulfillment incident INC-2025-041 carries a second-order finding: the alert rule for queue depth existed but was scoped to a metric the site did not emit. The reviewer's reading of incident INC-2025-041 is that an operational workaround masked the defect for longer than the incident record suggests. resource leak incident INC-2025-041 was raised again with the owning engineer after the initial interview, and the position did not change.

Schedule O - Robot variant, module revision and bench rig coverage (TDR-2025-14)

Robot variant	Module	Firmware revision	Sites deployed	Bench rig coverage	Revision status
HR-240 tote mover	safety scanner	1.2.4	9	neither rig	pre-production
HR-240 tote mover	wheel encoder set	4.16.8	8	neither rig	one revision behind
HR-200 tote mover	charge contact assembly	3.2.8	10	rig two only	superseded
HR-320 pallet mover	compute module	3.10.6	13	both rigs	pre-production
HR-300 pallet mover	network radio	4.12.4	5	both rigs	current
HR-320 pallet mover	wheel encoder set	3.5.0	13	rig two only	pre-production
HR-300 pallet mover	safety scanner	1.12.4	11	both rigs	superseded
HR-200 tote mover	gripper coprocessor	4.4.2	9	rig two only	current
HR-320 pallet mover	load cell deck	4.13.7	7	both rigs	pre-production
HR-300 pallet mover	network radio	4.16.6	6	neither rig	current
HR-240 tote mover	compute module	3.16.5	14	neither rig	superseded
HR-410 manipulator	wheel encoder set	4.15.6	8	both rigs	pre-production
HR-410 manipulator	wheel encoder set	2.8.7	7	neither rig	current
HR-410 manipulator	E-stop harness	1.12.8	8	rig two only	current
HR-410 manipulator	network radio	4.15.3	14	rig one only	superseded
HR-200 tote mover	depth camera	4.9.5	11	rig two only	pre-production
HR-240 tote mover	safety scanner	2.4.3	7	neither rig	superseded
HR-200 tote mover	charge contact assembly	2.15.6	9	rig two only	pre-production
HR-300 pallet mover	network radio	4.8.3	14	neither rig	superseded
HR-320 pallet mover	network radio	3.0.5	6	rig two only	one revision behind
HR-200 tote mover	network radio	4.12.6	5	neither rig	superseded
HR-410 manipulator	safety scanner	1.3.4	6	rig one only	current

5. Safety-relevant incidents and protective-stop behaviour - part 2 of 2

Because the 5. Safety-relevant incidents and protective-stop behaviour does not fit one page, its remainder is set out below under the same references. The sequence is unbroken either side of it.

Treat the conventions declared for the 5. Safety-relevant incidents and protective-stop behaviour as governing here too; this part introduces none of its own. The break is editorial.

Schedule O - Robot variant, module revision and bench rig coverage (TDR-2025-14) (continued, part 2 of 2)

Robot variant	Module	Firmware revision	Sites deployed	Bench rig coverage	Revision status
HR-200 tote mover	inertial measurement unit	1.13.2	4	rig one only	one revision behind
HR-200 tote mover	wheel encoder set	3.8.3	7	rig one only	current
HR-320 pallet mover	inertial measurement unit	2.14.7	5	rig one only	superseded
HR-320 pallet mover	inertial measurement unit	1.14.1	4	rig one only	one revision behind
HR-410 manipulator	safety scanner	4.4.0	13	rig two only	superseded
HR-300 pallet mover	charge contact assembly	2.16.1	8	rig one only	superseded
HR-240 tote mover	navigation lidar	4.16.6	11	rig one only	one revision behind
HR-240 tote mover	load cell deck	3.18.7	7	neither rig	current
HR-200 tote mover	navigation lidar	1.3.5	11	neither rig	one revision behind
HR-410 manipulator	compute module	2.9.1	7	rig two only	one revision behind
HR-410 manipulator	depth camera	4.6.5	9	rig two only	current
HR-240 tote mover	charge contact assembly	3.4.5	13	rig two only	current
HR-300 pallet mover	safety scanner	2.3.6	4	rig two only	pre-production
HR-320 pallet mover	navigation lidar	3.1.2	2	rig one only	current
HR-300 pallet mover	navigation lidar	1.17.5	14	rig one only	one revision behind
HR-200 tote mover	network radio	2.8.3	10	rig one only	pre-production
HR-320 pallet mover	E-stop harness	1.9.7	10	rig one only	current
HR-410 manipulator	navigation lidar	2.6.8	5	rig one only	current
HR-200 tote mover	inertial measurement unit	3.15.6	11	rig one only	one revision behind
HR-240 tote mover	safety scanner	4.10.3	3	rig two only	superseded
HR-300 pallet mover	load cell deck	3.15.7	13	rig one only	pre-production
HR-320 pallet mover	network radio	3.6.6	12	both rigs	one revision behind

6. Verification method and evidence behind each control

The sites that have moved to opportunity charging is the exposed population for hazard HZ-02 in the register. Register entry HZ-02 has produced no field event in the reviewed period, which is the strongest available test of the control. Evidence of verification at Serious injury hazard HZ-02 consists of site-specific commissioning test at each mapped blind corner. Serious injury hazard HZ-02 would move from a concern to a verified defect if one further measurement were supplied.

Remote-exposure hazard HZ-15 has produced no event, though the control has activated repeatedly; that is worth stating alongside the assessed exposure. Verification of HZ-15 was by site commissioning check plus a fire drill observation at two sites. Register entry HZ-15 was scoped to every site with a mezzanine or ramp, and the reviewer checked the scoping against the fleet record. Remote-exposure hazard HZ-15 is the reason one section of TDR-2025-14 concludes narrowly rather than generally.

Hazard HZ-16 in the register produced two near-miss reports from site staff, which is the strongest available test of the control. What the register does not say about hazard HZ-16 in the register, and the reviewer does, is that the ring pause defect in July 2025 did not produce a mixed-version robot. The control applied at hazard HZ-16 in the register is atomic update with rollback and a version consistency check before mission acceptance. hazard HZ-16 is the reason one section of TDR-2025-14 concludes narrowly rather than generally.

The sites that have moved to opportunity charging is the exposed population for hazard HZ-09 in the register. Evidence of verification at HZ-09 consists of ramp testing on a wet and a dry surface at two sites. HZ-09 has produced no event, though the control has activated repeatedly; that is worth stating alongside the assessed exposure. hazard HZ-09 is the reason one section of TDR-2025-14 concludes narrowly rather than generally.

Method of verification on Occasional-exposure hazard HZ-12: witnessed reverse test plus a review of the dock layout at four sites. Occasional-exposure hazard HZ-12 depends for its mitigation on reverse motion audible warning plus a rear sensing field. Minor injury hazard HZ-12 covers the case where operator is struck by a robot reversing out of a charge dock. register entry HZ-12 would be a candidate for the first post-close engineering review under any reasonable plan.

Occasional-exposure hazard HZ-22 is not clean: the tested duration is shorter than the five-hour outage seen in April 2026. Hazard HZ-22 has never activated in the field and is evidenced only on the bench; that is worth stating alongside the assessed exposure. Hazard HZ-22 is addressed by local yield behaviour with a conservative speed limit and a timed stop, and the reviewer traced that control into the design record. register entry HZ-22 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

Schedule M - Remediation commitment tracker with evidence of delivery (TDR-2025-14)

Commitment	Source	Owner	Committed date	Status	Evidence inspected
give every corridor reservation a lease with an expiry and add a site-level queue depth alert	INC-2025-041	Owen Brantley	2025-05-30	landed	release record inspected
re-derive the time offset on every reconnect and add the affected variant to the second bench ri	INC-2025-047	Yuki Harada	2025-06-15	partly landed	configuration export inspected
release charge windows on a shift-end event and reconcile the calendar nightly	INC-2025-052	Nadia Semenova	2025-06-30	landed	test added and observed
key suppression on the tenant and identifier pair and backfill the affected order records	INC-2025-058	Tomas Villareal	2025-07-15	landed	interview only
make the allocator consult charger availability before	INC-2025-063	Marta Kellerman	2025-07-31	landed	no evidence located

Commitment	Source	Owner	Committed date	Status	Evidence inspected
excluding a robot					
add a leadership and under-replicated partition check to the broker upgrade runbook	INC-2025-071	Erik Palmgren	2025-07-31	landed	runbook change inspected
invalidate the roster cache on a fleet membership event and add a session-level regression test	INC-2025-078	Anjali Sekhon	2025-08-31	landed	release record inspected
persist the pause state and require a design note for any change to the release state machine	INC-2025-084	Erik Palmgren	2025-08-15	landed	no evidence located
add the connector torque check to incoming inspection and retrofit the affected harness batch	INC-2025-090	Bettina Vogel	2025-09-30	partly landed	interview only
replace the mailbox fallback with a secondary responder and import rotations from the scheduling	INC-2025-096	Wanda Njoku	2025-10-15	partly landed	interview only
add a deterministic tie-break and return the pinch-point scenario to the nightly suite	INC-2025-103	Priya Raghunathan	2025-10-31	partly landed	configuration export inspected
fail the load on an incomplete partition and reprocess the affected days	INC-2025-110	Farida Cisse	2025-10-31	landed	runbook change inspected
redact authorization headers at the logging layer and cut build log retention to fourteen days	INC-2025-118	Sonia Krall	2025-10-31	partly landed	test added and observed
annotate staleness on every read path and add a rebalance-under-load test to the platform suite	INC-2025-124	Marta Kellerman	2025-11-30	landed	no evidence located
restrict export to published revisions and shorten the publication path so the workaround is unn	INC-2025-131	Lucia Ferrante	2025-12-15	partly landed	commit inspected
split control traffic onto a dedicated topic with its own quota and revisit the topic inventory	INC-2025-139	Erik Palmgren	2026-01-31	partly landed	release record inspected
fold the reset into the re-run command and make double-counting detectable in the ledger	INC-2025-146	Nadia Semenova	2025-12-31	landed	commit inspected
apply the lifecycle rule through the configuration baseline and delete the over-retained clips	INC-2025-152	Anjali Sekhon	2026-01-31	partly landed	runbook change inspected
replace the fallback matcher and re-survey the affected	INC-2026-004	Rahul Menon	2026-06-30	not landed	runbook change inspected

Commitment	Source	Owner	Committed date	Status	Evidence inspected
aisles					
require a calibration record before returning a manipulator robot to service	INC-2026-009	Ines Duarte	2026-04-15	landed	test added and observed
move precondition validation to apply time and add an offline apply case to the platform suite	INC-2026-015	Sam Oyelaran	2026-03-31	landed	commit inspected
pin the full build configuration and add a determinism check to the harness	INC-2026-021	Colin Achebe	2026-05-31	not landed	commit inspected
group on the signal and block closure while a constituent alert remains active	INC-2026-028	Wanda Njoku	2026-04-30	landed	commit inspected
measure closing rate where the sensing permits and re-run the protective distance calculation	INC-2026-033	Bettina Vogel	2026-08-31	not landed	interview only
serialise cancellation against assignment and add an hourly orphan sweep	INC-2026-039	Grace Lindqvist	2026-05-31	landed	no evidence located
aggregate buffer pressure to a site alert and extend the buffer to twelve hours on the current h	INC-2026-045	Sam Oyelaran	2026-06-30	partly landed	release record inspected
check every template against the baseline and re-converge the four forked templates	INC-2026-051	Marcus Duplessis	2026-06-15	partly landed	release record inspected
refit the estimator on opportunity-charge data and add an imbalance alert independent of state o	INC-2026-057	Peter Halvorsen	2026-09-30	not landed	release record inspected
require a justification and a second approver within one working day for every emergency release	INC-2026-062	Colin Achebe	2026-08-31	not landed	test added and observed
add a slip detector that flags the condition and issue a site advisory on cleaning windows	INC-2026-068	Yuki Harada	2026-09-15	not landed	configuration export inspected
Partition the fleet allocator write path	TD-01	engineering leadership	2027-03-31	in design	register entry only
Replace the unmaintained scan-matching library in the localization fallback	TD-02	engineering leadership	2026-12-31	committed	register entry only
Retire the vendored geometry library in costmap fusion	TD-03	engineering leadership	2027-06-30	backlog	register entry only
Raise robot software line coverage above sixty per cent on the perception path	TD-04	engineering leadership	2027-03-31	in progress	register entry only
Bring the second bench rig to the current sensor configuration	TD-05	engineering leadership	2026-09-30	committed	register entry only

Commitment	Source	Owner	Committed date	Status	Evidence inspected
Make the nightly simulation suite blocking	TD-06	engineering leadership	2026-11-30	blocked	register entry only

7. Safety-related engineering work not yet funded

TD-01 on Fleet Allocator carries a 2027-03-31 target and a status of in design. The load-test series identifies this as the single binding scalability constraint is the point the reviewer would put to management on programme item TD-01. Fleet Allocator item TD-01 is the item to partition the fleet allocator write path, classified under scalability. programme item TD-01 would move from a concern to a verified defect if one further measurement were supplied.

TD-22 on Payload Scale Interface is funded only if the team's headcount plan is met in the engineering plan the reviewer was shown. TD-22 on Payload Scale Interface carries an estimate of 0.25 engineer-quarters; the costed figure is $42,000. The register shows register item TD-22 as backlog, due 2027-01-31. register item TD-22 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

2026-09-30 is the target for IP hygiene item TD-24, whose status is committed. Programme item TD-24 carries an estimate of 0.25 engineer-quarters; the costed figure is $28,000. Funded in the current financial year is the funding position on register item TD-24. register item TD-24 has no single accountable owner, and that is the finding the reviewer would press hardest.

Whether delivery item TD-14 happens depends on funding, and it is funded from the second quarter onward. The business consequence recorded against register item TD-14 is that it bench queueing is the largest term in firmware lead time. In the programme programme item TD-14 is in parallel with the release-process changes. register item TD-14 was raised again with the owning engineer after the initial interview, and the position did not change.

Register item TD-17 carries a 2026-11-30 target and a status of in design. TD-17 on OTA Updater is sized at 0.5 engineer-quarters, or $72,000 at the company's loaded rate. TD-17 on OTA Updater is unfunded and dependent on capital rather than engineering effort; the register status and the funding position are not the same thing. programme item TD-17 was checked against the company's own register, which had not been updated since the change landed.

Effort on Costmap Fusion item TD-03 is estimated at 1 engineer-quarter, costed at $128,000. Funding for Costmap Fusion item TD-03 is partly funded, with the second phase unbudgeted, and the register does not say so. The dependency structure places support item TD-03 behind a platform change that has its own target date. programme item TD-03 is treated as material to the investment case rather than as a routine engineering item.

Whether TD-07 on Simulation Harness happens depends on funding, and it is funded in the current financial year. Status on register item TD-07 is overdue; the target is 2026-05-31. Simulation Harness item TD-07 covers the work to pin the full simulation build configuration and enable a determinism check. TD-07 on Simulation Harness was checked against the company's own register, which had not been updated since the change landed.

$144,000 is the costed estimate for TD-12, based on 1 engineer-quarter. The reviewer's qualification on register item TD-12 is that the site alert landed first; the buffer extension needs a storage change on the compute module. Item Edge Gateway Agent item TD-12 in the debt register is to extend the edge buffer to twelve hours, and it bears on Edge Gateway Agent. TD-12 was one of two items where TDR-2025-14 could not obtain the underlying data within the engagement.

TD-16 on Localization Engine cannot start earlier than it does because it is after the determinism work in the simulation harness. 1 engineer-quarter is the estimate on Localization Engine item TD-16, which converts to $126,000. Whether TD-16 on Localization Engine happens depends on funding, and it is partly funded, with the second phase unbudgeted. register item TD-16 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

The dependency structure places programme item TD-02 in parallel with the release-process changes. Exposure carried by IP and support item TD-02 is that it strong-copyleft exposure in shipped robot software plus an unsupported dependency. TD-02 on Localization Engine carries an estimate of 1.5 engineer-quarters; the costed figure is $185,000. IP and support item TD-02 would be a candidate for the first post-close engineering review under any reasonable plan.

Sequencing of TD-15 matters: it is last in a three-item chain, so its date is the least reliable on the register. The reviewer's qualification on TD-15 on Alarm Router is that raised in the 2026 assessment and accepted for remediation. Register item TD-15 covers the work to add per-region credentials for the paging provider. TD-15 on Alarm Router is the entry that changed most between the first and second document requests behind TDR-2025-14.

Reading programme item TD-09 against the funded plan, the upgrade requires a maintenance window at every site. The estimate for Event Bus item TD-09 is 1.5 engineer-quarters and $196,000. Reading the programme plan, register item TD-09 is in parallel with the release-process changes. support item TD-09 is the clearest instance in TDR-2025-14 of engineering practice being better than the documentation suggests.

On register item TD-13 the reviewer records that carries the same dependency as the copyleft removal. Programme item TD-13 is funded and already staffed; the register status and the funding position are not the same thing. The register entry TD-13 on Localization Engine reads: replace the fallback scan matcher with the current implementation, in the performance category. programme item TD-13 was checked against the company's own register, which had not been updated since the change landed.

8. Safety-rated components and third-party assessment

In Order Broker the component http-router is used cloud service routing, which is the fact the licence analysis rests on. OSS-022 covers http-router at version 1.9.4, used cloud service routing. Policy compliance for http-router rests on its approval entry, which is approved 2024-01-09. http-router was tested rather than accepted, and the test is described in the working papers behind TDR-2025-14.

Tsdb-client was assessed for support as well as licence risk, and it carries no identified risk beyond ordinary version currency. Tsdb-client reaches the customer only through Telemetry Ingest, and the reviewer confirmed there is no second path. Attribution present in the cloud notice file is the reviewer's observation on tsdb-client. tsdb-client was inspected in the repository rather than described from the ticket, which is why the account is specific.

Scanmatch-legacy is licensed under GPL-2.0, with copyleft exposure recorded as strong. Scanmatch-legacy carries a practical obligation: to exclude a particular build configuration from the shipped artefact. Provenance of scanmatch-legacy is vendored source copied from a project archive in 2022. scanmatch-legacy is the reason one section of TDR-2025-14 concludes narrowly rather than generally.

No notice obligations beyond attribution is the reviewer's observation on grpc-transport. Grpc-transport affects Event Bus; no other component reviewed depends on it. What the company must actually do about grpc-transport is to share alike any derived asset it distributes. grpc-transport is included because a reader of TDR-2025-14 should be able to disagree with the inference without repeating the work.

The inventory entry OSS-006 is chartkit 4.8.2, operator console visualisations. To publish the modified files it distributes is what chartkit requires in practice. Tracing chartkit to its source gives public registry; upstream archived in 2025. chartkit is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

On provenance, sqlite-embedded shows public registry, signature verified. Sqlite-embedded was assessed for support as well as licence risk, and it carries a single-maintainer dependency with no succession plan. In Edge Gateway Agent the component sqlite-embedded is used on-robot local state, which is the fact the licence analysis rests on. sqlite-embedded was reviewed against the position twelve months earlier, and nothing measurable has changed.

Compliance with the jwt-verify terms requires to reproduce the notice text in the delivered documentation. Jwt-verify was obtained as follows: public registry, signature verified. On jwt-verify the reviewer records that attribution present. jwt-verify carries a follow-up request that was satisfied after the draft of TDR-2025-14 was circulated.

In Billing Usage Meter the component csv-export is used billing and usage exports, which is the fact the licence analysis rests on. Csv-export carries this qualification: attribution present. To reproduce the notice text in the delivered documentation is what csv-export requires in practice. csv-export is the clearest instance in TDR-2025-14 of a control that exists on paper and is weaker in practice.

Public registry, signature verified is the provenance recorded for image-signing. Image-signing reaches the customer only through OTA Updater, and the reviewer confirmed there is no second path. Image-signing carries

Apache-2.0 terms, classified no for copyleft purposes. image-signing was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

What the company must actually do about opencv-subset is to attribute the component in the customer-facing notice file. The inventory entry OSS-011 is opencv-subset 4.9.0, image preprocessing for grasp proposals. The non-licence risk on opencv-subset is a fork that must be maintained locally for as long as it is used. opencv-subset was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

The terms applying to physics-core-fork are Zlib; the copyleft classification is no. Approved with conditions 2023-04-18 is the approval entry for physics-core-fork. The open question on physics-core-fork is that the fork diverges by nineteen patches and no upstream merge path is maintained. physics-core-fork is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

9. Security findings with a safety consequence

Evidence behind finding SEC-2026-05 is that credential reuse confirmed from the secret store inventory, which the reviewer treats as moderate, because the demonstration was performed in a laboratory tenant rather than in production. As at the review date finding SEC-2026-05 was open and the closure entry reads committed 2026-09-30. Reading Alarm Router finding SEC-2026-05, the defect is that alarm router delivery provider credential shared across regions; the discovery date is 2026-04-16. SEC-2026-05 matters because one leaked credential affects paging for every region. Medium-severity finding SEC-2026-05 was reviewed against the position twelve months earlier, and nothing measurable has changed.

SEC-2026-07 against Hardware-in-the-Loop Rig matters because an unpatched laboratory host sits on the engineering network. The tester scored finding SEC-2026-07 at 6.7 and labelled it Medium. Finding SEC-2026-07 produces this judgement: the underlying pattern appears in two other components that the tester did not reach. 2026-04-17 is the discovery date of finding SEC-2026-07, which finds that hardware-in-the-loop rig controller runs an unsupported operating system. assessment finding SEC-2026-07 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

For finding SEC-2025-07 the record shows that the secret was recovered from a template export, which is strong evidence and is graded separately from severity. Tracking Medium-severity finding SEC-2025-07 to closure gives a status of remediated and a date of 2026-01-16. At 5.9 the score on SEC-2025-07 places it in the Medium group. Assessment finding SEC-2025-07 matters because the same secret was reusable at any site provisioned from that template. SEC-2025-07 is the reason one section of TDR-2025-14 concludes narrowly rather than generally.

On OTA Updater finding SEC-2025-09 the conclusion is that the compensating control is real but depends on a network boundary that a single misconfiguration removes. Residual exposure on assessment finding SEC-2025-09 reads medium; a two-person rule is designed but not enforced in the tool. In SEC-2025-09 against OTA Updater the exploitation path means a single key holder could sign an unreviewed firmware artefact. For finding SEC-2025-09 the record shows that key custody was evidenced by interview and the key ceremony record, which is limited, resting on a configuration read rather than on a demonstrated exploitation evidence and is graded separately from severity. finding SEC-2025-09 was reviewed against the position twelve months earlier, and nothing measurable has changed.

Rating for finding SEC-2026-18 is Low, band 3.4; the reviewer re-read the scoring rationale and accepted it. What Battery Management Service finding SEC-2026-18 enables is this: a diagnostic flood can slow pack telemetry reporting. Assessment finding SEC-2026-18 records that battery pack diagnostic interface has no rate limit, discovered on 2026-04-23. Low-severity finding SEC-2026-18 stands open; committed 2026-12-15 is what the record gives for closure. assessment finding SEC-2026-18 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Weighing Medium-severity finding SEC-2026-16 against the compensating controls, the severity is right and the ownership is unclear, since two teams each believe the other holds it. Costmap Fusion finding SEC-2026-16 was graded Medium, band 5.8, by Meridian Assurance engagement MA-2026-1. The practical consequence in SEC-2026-16 is that on-network access to a live perception diagnostic stream. On 2026-04-22 the tester recorded SEC-2026-16 against Costmap Fusion: costmap fusion diagnostic socket bound to all interfaces. SEC-2026-16 against Costmap Fusion would be a candidate for the first post-close engineering review under any reasonable plan.

2026-06-18 is the closure position on finding SEC-2026-14, which the ledger records as remediated. The evidence for finding SEC-2026-14 is moderate: action performed and the audit trail checked. Finding SEC-2026-14 produces this judgement: the severity is right and the ownership is unclear, since two teams each believe the other holds it. Finding SEC-2026-14 translates into the following exposure: a manual corridor override cannot be attributed after the fact. SEC-2026-14 was inspected in the repository rather than described from the ticket, which is why the account is specific.

SEC-2026-15 is accepted, with a closure date of not planned. At 3.0 the score on SEC-2026-15 against Site Provisioning Service places it in the Low group. What SEC-2026-15 enables is this: operator confusion rather than a direct security impact. After the fix, finding SEC-2026-15 carries residual risk described as low; management judged the operational cost of enforcement higher than the risk. SEC-2026-15 would move from a concern to a verified defect if one further measurement were supplied.

Severity on Telemetry Ingest finding SEC-2025-05 is High, scored 7.8 by the assessor. In SEC-2025-05 against Telemetry Ingest the exploitation path means a forged agent could inject telemetry attributed to a real robot. Telemetry ingest accepted a robot identity without checking device attestation is the substance of SEC-2025-05 against Telemetry Ingest, raised on 2025-09-19. Finding SEC-2025-05 is supported by the fact that the tester submitted synthetic telemetry from an unenrolled host; evidence strength is strong. SEC-2025-05 against Telemetry Ingest was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Release Orchestrator finding SEC-2026-01 is open, with a closure date of committed 2026-08-31. After the fix, Release Orchestrator finding SEC-2026-01 carries residual risk described as high until the second-approver control lands. Emergency release path permits a single approver is the substance of High-severity finding SEC-2026-01, raised on 2026-04-14. Read alongside the remediation record, Release Orchestrator finding SEC-2026-01 shows that the underlying pattern appears in two other components that the tester did not reach. finding SEC-2026-01 illustrates why TDR-2025-14 separates the severity of a finding from the confidence behind it.

SEC-2026-19 against Localization Engine translates into the following exposure: a stolen compute module discloses a customer's warehouse layout. SEC-2026-19 is supported by the fact that cache read from a removed module; evidence strength is strong. Localization Engine finding SEC-2026-19 was opened on 2026-04-23 and states that localization map cache stored unencrypted on the robot. Read alongside the remediation record, SEC-2026-19 shows that the remediation is adequate and the time to remediate is not. assessment finding SEC-2026-19 was cross-checked against the engineering record supplied for TDR-2025-14, and the two agree.

10. Open safety qualifications carried forward

The reviewer has resisted the temptation to tidy the present report: the open items are open because the evidence is not there yet. Pinehurst Systems Assurance will not treat an open item in the present report as closed on a verbal assurance, and has said so to the engineering leadership. On the evidence available by 2025-10-07, the report at TDR-2025-14 concludes that there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. Pinehurst Systems Assurance concludes, on the material examined, that there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. The report at TDR-2025-14 does not restate the company's own reporting where the reviewer was able to verify the underlying data instead.

Not every finding in the report at TDR-2025-14 is accepted by management: 4 are declined, and the declinatures are reproduced rather than argued away. 4 declined items are carried in report TDR-2025-14, of which two lack any documented compensating control. Work on TDR-2025-14 ran over 5 weeks and included 15 interviews and read-only access to the engineering repositories. The working method behind the present report was to start from the incident record and work back to the design, rather than the other way round. The present report was prepared on the footing that a reader may be an engineer or may not, and defines terms where the distinction matters.

Report TDR-2025-14 grades evidence as strong, moderate or limited, and a limited grade is not upgraded because the consequence would be serious. Grading in the report at TDR-2025-14 is deliberately coarse, because a finer scale would imply a precision the underlying evidence does not carry. Interview evidence in the report at TDR-2025-14 came from 15 sessions across 5 teams, each minuted and circulated for correction. Interviews for the present report were

conducted with the owning engineer present rather than with a manager speaking on their behalf. Report TDR-2025-14 is one input to a diligence process and is not a substitute for the company's own engineering judgement.

TDR-2025-14 proposes 6 metrics for the post-close monitoring pack, each already produced by an existing system. The monitoring pack the present report proposes would have surfaced two of the incidents examined here before a customer noticed. Selection for report TDR-2025-14 favoured components that appear in the incident record, which biases the sample towards known trouble and is stated as such. Report TDR-2025-14 sampled 14 services from an inventory of 34, weighted towards those on the critical control path. Pinehurst Systems Assurance has costed remediation at the company's own loaded engineering rate rather than at a market rate.

11. Conclusions on the safety position

Stated without qualification for once, the conclusion of the present report is that there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. K. Vandermeer would summarise the present report in one sentence: there is no finding that would prevent the financing, and there are three that should be reflected in the post-close plan with named owners. Sampling in the present report covered 14 of the 34 catalogued services and 4 of the 18 live sites. The sample behind report TDR-2025-14 is 14 services and 4 sites; the selection was the reviewer's, not the company's. Pinehurst Systems Assurance was given read access without restriction on scope within the agreed subject matter.

For the transaction, TDR-2025-14 identifies 2 matters the reviewer would raise before closing and 4 for the post-close engineering plan. The present report suggests one governance change rather than a price adjustment: a quarterly technical review with the lead investor's observer present. The open items in the report at TDR-2025-14 are grouped by what would resolve them rather than by which team owns them, because several cross team boundaries. 4 matters remain open at the date of this report, and none of them is closed by the passage of time alone. K. Vandermeer interviewed 15 people for the present report; where an account was uncorroborated it is marked as such.

The pack the present report proposes is deliberately short, because a longer one would be produced and not read. Report TDR-2025-14 recommends that the remediation commitment register be reported to the board quarterly with evidence attached, not just status. Where TDR-2025-14 reports a supplier's test result, the reviewer read the certificate and did not witness the test. Nothing in the present report rests on a test the reviewer designed; the tests were the company's, and the reviewer read the results and the method. TDR-2025-14 deliberately preserves detail that a summary would drop, because the detail is where the qualifications live.